

CMAN-8

Given Number of Neurons $N = 1024$

Mean firing rate per neuron $f = 50 \text{ Hz}$

AER packet

= 10 bit neuron address + 6 bit time stamp + 4 bit framing/parity

= 20 bits total

TASKS:

1. $R = N \times f$ = Mean Aggregate spike rate

$$R = 1024 \times 50$$

$$R = 51,200 \text{ spikes/second}$$

2. Mean Required AER Bandwidth

$$B = R \times 20$$

$$B = 51,200 \times 20$$

$$B = 1,024,000 \text{ bits/second}$$

$$B = 1.024 \text{ Mbits/s}$$

3. For the interface Comparison

Interface	Maximum throughput	Can sustain Mean rate (1.024 Mbits/s)
1^2C	$\leq 3.4 \text{ Mbits/s}$	Yes
SPD	$\leq 50 \text{ Mbits/s}$	Yes
AXI4-Lite	$\approx 100 \text{ Mbits/s}$	Yes

Here all the three interfaces can sustain the mean AER traffic rate of 1.024 Mbit/sec

Here the lowest-complexity interface among all the three is I^2C compared to SPI and AXI4-like as they are more complex to implement. I^2C uses only 2 wires: SDA & SCL.

4. Burst Analysis

25% of 1024 neurons fire within a 1ms window

$$\begin{aligned}\text{Neurons firing} &= 0.25 \times 1024 \\ &= 256 \text{ Neurons}\end{aligned}$$

Peak Instantaneous Bandwidth

$$\begin{aligned}B_{\text{peak}} &= 5,120 \text{ bits}/0.001 \text{ s} \\ &= 5.12 \text{ Mbit/sec}\end{aligned}$$

$$\text{Burst to Mean ratio} = \frac{5.12}{1.024}$$

$\Rightarrow 5:1$ 5x more the mean bandwidth

The chosen interface I^2C 3.4 Mbit/sec cannot absorb the burst $5.12 \text{ Mbit/s} > 3.4 \text{ Mbit/s}$

Buffering will be required.

$$\begin{aligned}\text{Excess data during burst} &= (5.12 - 3.4) \times 10^6 \times 0.001 \text{ s} \\ &= 17,200 \text{ bits} \approx 86 \text{ packets}\end{aligned}$$

This is equal to 86 packets of buffer depth required
Ax14 lite & SPI absorb burst directly without buffering

5. Frame-Based Comparison

→ 1024 neurons sampled every 1ms, 1bit per neuron

→ Bits per frame = 1024 bits

→ Frames per second = 1,000

→ B-frame = 1024×1000

= 1,024,000 bits/s.

= 1.024 mbits/s

AER - to-frame bandwidth ratio at $f = 50 \text{ Hz}$

$$\frac{\text{B-AER}}{\text{B-frame}} = \frac{1.024}{1.024} = 1:1$$

Crossover firing rate

Set B-AER = B-frame

$$N \times f_{\text{-crossover}} \times 20 = N \times 1000$$

$$20 \times f_{\text{-crossover}} \times 20 = N \times 1000$$

$$20 \times f_{\text{-crossover}} = 1000$$

$$f_{\text{-crossover}} = 50 \text{ Hz}$$

AER is right choice with mean firing rates are below at 50 Hz because at sparser activity the event-driven bandwidth is lower than the fixed frame-based cost, above 50 Hz. frame-based readout becomes more bandwidth-efficient.